

Math 421

Friday, October 24

Announcements

- Homework 6 Due **today**

lim sup **and** lim inf

Def. Let (s_n) be a sequence in $\mathbb{R}$. We define

$$\liminf s_n = u = \lim_{N \rightarrow \infty} u_N = \lim_{N \rightarrow \infty} \inf\{s_n : n > N\}$$

and

$$\limsup s_n = v = \lim_{N \rightarrow \infty} v_N = \lim_{N \rightarrow \infty} \sup\{s_n : n > N\}$$

Examples.

Activity

Let (s_n) be the sequence defined by $s_n = \sin\left(\frac{3\pi}{2}\right)$.

1. Determine $S_N = \{s_n : n > N\}$ for some explicit values of N .
2. Determine $u_N = \inf\{s_n : n > N\}$ and $v_N = \sup\{s_n : n > N\}$.
3. Compute $\liminf s_n$ and $\limsup s_n$.
4. How are $\liminf s_n$ and $\limsup s_n$ related to the set of subsequential limits?

Discussion Questions

Convince yourselves the following properties are true:

❖ $\limsup s_n \leq \sup\{s_n : n \in \mathbb{N}\} \text{ \& } \liminf s_n \geq \inf\{s_n : n \in \mathbb{N}\}$

❖ $\liminf s_n \leq \limsup s_n$

❖ If $\lim s_n$ exists then

❖ $u_N \leq \lim s_n \leq v_N$ for all N

❖ $\liminf s_n \leq \lim s_n \leq \limsup s_n$

$$\lim s_n = \liminf s_n = \limsup s_n$$

Theorem 10.7: Let (s_n) be a sequence in $\mathbb{R}$, (s_n) has a limit if and only if $\liminf s_n = \limsup s_n$.

Discussion.

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Proof ($\Rightarrow$). Infinite limits.

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Proof ($\Rightarrow$). Finite limits.

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Proof ($\Leftarrow$). Finite limits.

Theorem 11.7

Theorem. Let (s_n) be any sequence. There exists a subsequence whose limit is $\limsup s_n$, and there exists a subsequence whose limit is $\liminf s_n$.

Discussion.

Theorem 11.8

Thm. Let (s_n) be any sequence in $\mathbb{R}$, and let S denote the set of subsequential limits of (s_n) .

1. S is nonempty.
2. $\sup S = \limsup s_n$ and $\inf S = \liminf s_n$.
3. $\lim s_n$ exists if and only if S has exactly one element, namely $\lim s_n$.

❖ Convince yourselves this theorem is true. Parts 1 & 3 follow from previous theorems, Part 2 requires more proof – to convince yourselves just think of some examples.

Activity

Give sequences that exhibit the following properties:

- ❖ The set $\{n : s_n > \limsup s_n\}$ can be infinite. **Hint:** We have already seen a sequence that has this property.
- ❖ The set $\{n : s_n < \liminf s_n\}$ can be infinite.
- ❖ If $\liminf s_n < \limsup s_n$, the set
$$\{n : \liminf s_n \leq s_n \leq \limsup s_n\}$$
can be empty. **Hint:** Try to sketch out what this property is saying, what would need to be true about a sequence that satisfies this property?

Chapter 2: Sequences

Section 10 (Part 3): Cauchy Sequences

Cauchy Sequences

Def: A sequence (s_n) of real numbers is called a Cauchy sequence if for each $\epsilon > 0$ there exists a number such that

Q: How does this definition compare to the definition of a convergent sequence?

Convergent $\Rightarrow$ Cauchy

Thm. Convergent sequences are Cauchy sequences.

Activity: Prove this Lemma.

- 1 What are you given?
- 2 What are you trying to show?
- 3 Sketch the details.

Cauchy $\Rightarrow$ Bounded

Thm. Cauchy sequences are bounded

Activity: Prove this Lemma.

- 1 What are you given?
- 2 What are you trying to show?
- 3 Sketch the details.

Hint: Follows similarly to proving convergent sequences are bounded.

Convergent $\Leftrightarrow$ Cauchy

Thm. A sequence is a convergent sequence if and only if it is a Cauchy sequence.

Q: Why might this theorem be useful?

Recall – From Ross §14

Consider the series given by $\sum_{n=1}^{\infty} \frac{n}{n^2+3}$. Find the first 4 terms in its sequence of partial sums.

What does it mean for an infinite series to converge?

What does it mean for an infinite series to be absolutely convergent?

Cauchy Criterion

Def. We say a series $\sum a_n$ satisfies the Cauchy criterion if its sequence of partial sums is a Cauchy sequence.

Thm 14.4. A series converges if and only if it satisfies the Cauchy criterion.

Activity

Cor 14.5. If a series $\sum a_n$ converges, then $\lim a_n = 0$.

Prove this statement using Thm 14.4.

Comparison Test

Thm. Let $\sum a_n$ be a series where $a_n \geq 0$ for all n . If $\sum a_n$ converges and $|b_n| \leq a_n$ for all n , then $\sum b_n$ converges.

Discussion.

Discussion Question

Think back over our discussions about sequences. How many different ways can you think of to prove that a given sequence does not converge?